Chapter 4. Classification: Logistic Regression, Softmax and SVM

Logistic regression, multiclass classification, support vector machines

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1 Odds, Logit and the Logistic Model

Motivation. Linear log-odds give the logistic model: a sigmoid of a linear score.

Odds $o = p/(1-p)$ ($p = 0.8$: odds 4) and

$$\operatorname{logit}(p) = \log \frac{p}{1-p} \in \mathbb{R}, \qquad \sigma(z) = \frac{1}{1+e^{-z}} = \operatorname{logit}^{-1}(z).$$

Definition 4.1: logistic regression

With augmented input $\mathbf{x} = (1, x_1, \ldots, x_d)$ and $y \in \{0, 1\}$,

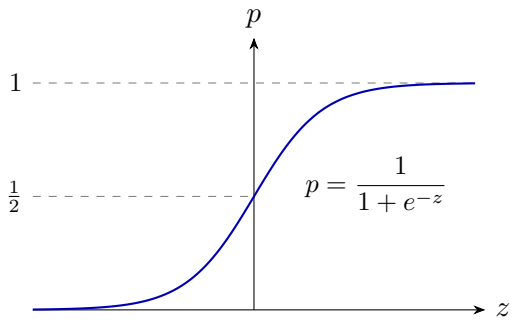
$$\log \frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}, \qquad P(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}).$$

Thresholding at τ gives the hyperplane $\mathbf{w}^\top \mathbf{x} = \log \frac{\tau}{1-\tau}$ (at $\tau = 0.5$: $\mathbf{w}^\top \mathbf{x} = 0$).

Theorem 4.2: sigmoid identities

$\sigma'(z) = \sigma(z)(1-\sigma(z)) \le \frac{1}{4}$, $1 - \sigma(z) = \sigma(-z)$, $\frac{d}{dz} \log(1+e^z) = \sigma(z)$.

Proof. $\sigma'(z) = e^{-z}/(1+e^{-z})^2 = \sigma(z) \cdot \frac{e^{-z}}{1+e^{-z}}$ and $\frac{e^{-z}}{1+e^{-z}} = \frac{1}{e^z+1} = \sigma(-z) = 1 - \sigma(z)$. The maximum of $s(1-s)$ on $[0,1]$ is $1/4$ at $s = 1/2$. ■



The logistic function, inverse of the logit $z = \log \frac{p}{1-p}$.

2 Likelihood, Gradient, Hessian and Convexity

Motivation. No closed form, but a PSD Hessian: the loss is convex and gradient methods are reliable.

For data $(\mathbf{x}_i, y_i)$ with $p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = p_i^{y_i}(1-p_i)^{1-y_i}$ and

$$J(\mathbf{w}) = -\sum_i \left[y_i \log p_i + (1-y_i) \log(1-p_i) \right] = \sum_i \left[\log(1+e^{z_i}) - y_i z_i \right], \quad z_i = \mathbf{w}^\top \mathbf{x}_i.$$

Theorem 4.3: logistic loss is convex

$\nabla J = \sum_i (p_i - y_i)\mathbf{x}_i = X^\top(\mathbf{p} - \mathbf{y})$ and $\nabla^2 J = \sum_i p_i(1-p_i)\mathbf{x}_i\mathbf{x}_i^\top = X^\top S X$, $S = \operatorname{diag}(p_i(1-p_i))$. Hence J is convex, and strictly convex if X has full column rank.

Proof. $\partial J/\partial z_i = \sigma(z_i) - y_i = p_i - y_i$, $\partial z_i/\partial \mathbf{w} = \mathbf{x}_i$, $\nabla p_i = p_i(1-p_i)\mathbf{x}_i$. Then $\mathbf{v}^\top X^\top S X \mathbf{v} = \sum_i p_i(1-p_i)(\mathbf{x}_i^\top \mathbf{v})^2 \ge 0$, positive when $X\mathbf{v} \neq \mathbf{0}$ since $p_i(1-p_i) > 0$. ■

Example 4.4: one gradient step

$\mathbf{x} = (1, 2)$, $y = 1$, $\mathbf{w} = \mathbf{0}$: $p = 1/2$, $\nabla J = (p - 1)\mathbf{x} = (-1/2, -1)$. With $\eta = 0.1$, $\mathbf{w} = (0.05, 0.1)$, score 0.25, $p \approx 0.562$ (up).

3 Labels in $\{\pm 1\}$, Logistic Loss and BCE

Motivation. With labels $\{\pm 1\}$ the loss depends on the margin yf, linking logistic regression to the SVM.

With $Y \in \{\pm 1\}$, $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$, $P(Y = \pm 1 \mid \mathbf{x}) = \sigma(\pm f)$, so $-\log P(Y \mid \mathbf{x}) = \ell(Yf)$ with the **logistic loss** $\ell(z) = \log(1 + e^{-z})$; the *margin* $Yf(\mathbf{x})$ is positive iff the sign is correct.

Theorem 4.5: three names for one objective

Let $P(-1) = (1, 0)$, $P(+1) = (0, 1)$, $\mu(r) = (\sigma(-r), \sigma(r))$, $\tilde{Y} = (1 + Y)/2 \in \{0, 1\}$ and $\mathrm{BCE}(s, t) = -[t \log s + (1 - t) \log(1 - s)]$. For every $(\mathbf{a}, b)$ and every i,

$$D_{\mathrm{KL}}(P(Y_i) \,\|\, \mu(f(\mathbf{x}_i))) = \mathrm{BCE}\left(\sigma(f(\mathbf{x}_i)), \tilde{Y}_i\right) = \ell(Y_i f(\mathbf{x}_i)).$$

Proof. One-hot KL $= -\log$(model probability of the true class). $Y = +1$: $-\log \sigma(r) = \ell(r)$, and BCE with $t = 1$ is $-\log s$. $Y = -1$: $-\log \sigma(-r) = \ell(-r)$, and BCE with $t = 0$ is $-\log(1 - s) = -\log \sigma(-r)$. ■

Theorem 4.6: ℓ is convex

$\ell'(z) = \sigma(z) - 1$ and $\ell''(z) = \sigma(z)\sigma(-z) > 0$.

3.1 Surrogate risk bounds

For $f = \mathrm{sign}(g)$ ($g = 0$ counted as an error), $\{f(X) \neq Y\} \subseteq \{Yg(X) \leq 0\}$, so for nonnegative nonincreasing ϕ, $\phi(Yg) \geq \phi(0)$ on errors and

$$P(f(X) \neq Y) \leq \frac{\mathbb{E}[\phi(Yg(X))]}{\phi(0)}.$$

Logistic: $\phi(0) = \log 2$, so $R_{\log} < \varepsilon \Rightarrow P(f(X) = Y) > 1 - \varepsilon/\log 2$. Hinge $\max(0, 1 - u)$ and exponential e^{-u} have $\phi(0) = 1$. A convex surrogate thus controls the 0–1 error.

4 Multiclass Classification and Softmax

Motivation. Softmax maps K scores to probabilities, with gradient "prediction minus target".

For Q classes, $W \in \mathbb{R}^{Q \times d}$, scores $s_k = \mathbf{w}_k^\top \mathbf{x}$ and

$$p_k(\mathbf{x}) = \mathrm{softmax}(\mathbf{s})_k = \frac{e^{s_k}}{\sum_r e^{s_r}}, \qquad J(W) = -\sum_{i,k} y_{ik} \log p_k(\mathbf{x}_i).$$

Scores $(2, 1, 0)$ give approximately $(0.665, 0.245, 0.090)$.

Theorem 4.7: softmax calculus

With $\mathrm{LSE}(\mathbf{s}) = \log \sum_r e^{s_r}$ and $\mathbf{p} = \mathrm{softmax}(\mathbf{s})$:

(i) $\mathrm{softmax}(\mathbf{s} + c\mathbf{1}) = \mathrm{softmax}(\mathbf{s})$, $\mathrm{LSE}(\mathbf{s} + c\mathbf{1}) = \mathrm{LSE}(\mathbf{s}) + c$; stable evaluation $\mathrm{LSE}(\mathbf{s}) = M + \log \sum e^{s_r - M}$, $M = \max s_r$.

(ii) $\nabla \,\mathrm{LSE} = \mathbf{p}$; Jacobian $\partial p_k / \partial s_m = p_k(\delta_{km} - p_m)$, i.e. $\nabla^2 \,\mathrm{LSE} = \mathrm{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top \succeq 0$.

(iii) $\nabla_{\mathbf{s}}(-\log p_y) = \mathbf{p} - \mathbf{e}_y$; for the whole data set $\nabla_W J = (P - Y)^\top X$ (rows: classes).

(iv) $\log \mathrm{softmax}(\mathbf{s}) = \mathbf{s} - \mathrm{LSE}(\mathbf{s})\mathbf{1}$ and log-softmax is idempotent: $\log \mathrm{softmax}(\log \mathrm{softmax}(\mathbf{s})) = \log \mathrm{softmax}(\mathbf{s})$.

(v) $\max_k s_k \le \mathrm{LSE}(\mathbf{s}) \le \max_k s_k + \log Q$.

Proof. (i) e^c cancels.

(ii) Quotient rule; $\mathbf{v}^\top(\mathrm{diag}\,\mathbf{p} - \mathbf{p}\mathbf{p}^\top)\mathbf{v} = \sum p_k v_k^2 - (\sum p_k v_k)^2$ is the variance of v_K with $K \sim \mathbf{p}$.

(iii) $-\log p_y = -s_y + \mathrm{LSE}(\mathbf{s})$.

(iv) With $\boldsymbol{\ell} = \log \mathrm{softmax}(\mathbf{s})$, $e^{\ell_j} = p_j$ so $\mathrm{LSE}(\boldsymbol{\ell}) = \log \sum p_j = 0$.

(v) $e^M \le \sum e^{s_r} \le Q e^M$. ■

Adding one vector to all rows of W leaves the softmax unchanged: W is identifiable only up to shifts.

5 Support Vector Machines

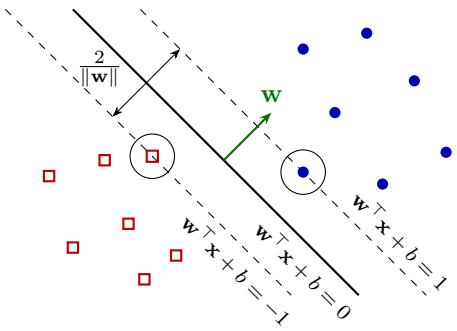
5.1 Geometry and the primal problem

$y_i \in \{\pm 1\}$, $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$. The distance from $\mathbf{x}_0$ to $\{\mathbf{w}^\top \mathbf{x} + b = 0\}$ is $|\mathbf{w}^\top \mathbf{x}_0 + b| / \|\mathbf{w}\|$ ($\mathbf{x}_0 - t\mathbf{w}/\|\mathbf{w}\|$ is on the plane iff $t = (\mathbf{w}^\top \mathbf{x}_0 + b)/\|\mathbf{w}\|$). $(\mathbf{w}, b)$ is scale-free, so normalize the closest functional margin to 1:

Definition 4.8: hard-margin SVM

$$\min_{\mathbf{w},b} \ \tfrac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \ge 1, \ \ i = 1, \dots, n.$$

The support planes $\mathbf{w}^\top \mathbf{x} + b = \pm 1$ lie at distance $1/\|\mathbf{w}\|$ (the margin) from the boundary; the slab width is $2/\|\mathbf{w}\|$.



Slab width $2/\|\mathbf{w}\|$; circled points are support vectors.

5.2 Lagrangian, max–min inequality, weak duality

Theorem 4.9: max–min inequality

For any $F : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, $\sup_x \inf_y F(x,y) \le \inf_y \sup_x F(x,y)$. Equality can fail: for $F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (rows x, columns y) the left side is 0 and the right side 1.

Proof. $\inf_{y'} F(x,y') \le F(x,y) \le \sup_{x'} F(x',y)$; take $\sup_x$ of the left end and $\inf_y$ of the right. ■

With $c_i = 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)$, $\alpha_i \ge 0$, $L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_i \alpha_i c_i$. As $\sup_{\alpha \ge 0} \alpha c$ is 0 ($c \le 0$) or $+\infty$ ($c > 0$), $p^\star = \inf_{\mathbf{w},b} \sup_{\boldsymbol{\alpha}} L$. The dual $d^\star = \sup_{\boldsymbol{\alpha}} \inf_{\mathbf{w},b} L$ satisfies **weak duality** $d^\star \le p^\star$; for the SVM (convex, strictly feasible) $d^\star = p^\star$.

5.3 The dual problem

Theorem 4.10: SVM dual

$\nabla_{\mathbf{w}} L = \mathbf{0} \Rightarrow \mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$; finiteness of $\inf_b L$ requires $\sum_i \alpha_i y_i = 0$. The dual is

$$\max_{\boldsymbol{\alpha} \ge \mathbf{0}} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0.$$

Complementary slackness $\alpha_i[1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)] = 0$: $\alpha_i > 0$ only for support vectors, and $b = y_i - \mathbf{w}^\top \mathbf{x}_i$ for any such i.

Proof. b enters L linearly with coefficient $-\sum \alpha_i y_i$; if nonzero, $\inf_b = -\infty$. Substituting $\mathbf{w}$: $\frac{1}{2}\|\mathbf{w}\|^2 - \|\mathbf{w}\|^2 + \sum \alpha_i = \sum \alpha_i - \frac{1}{2}\|\mathbf{w}\|^2$. ■

Example 4.11: two points

$\mathbf{x}_1 = (0,0)$, $y_1 = -1$; $\mathbf{x}_2 = (2,2)$, $y_2 = +1$. $\alpha_1 = \alpha_2 = a$, $\|\mathbf{x}_2\|^2 = 8$: dual $2a - 4a^2$, maximal at $a = 1/4$. Then $\mathbf{w} = (1/2, 1/2)$, $b = -1$, boundary $x^{(1)} + x^{(2)} = 2$, margin $\sqrt{2}$ (slab $2\sqrt{2}$); primal $\frac{1}{2}\|\mathbf{w}\|^2 = 1/4 =$ dual.

Remark 4.12: soft margin and kernels

Slacks: $\min \frac{1}{2}\|\mathbf{w}\|^2 + C\sum \xi_i$ s.t. $y_i f(\mathbf{x}_i) \ge 1 - \xi_i$, $\xi_i \ge 0$ (hinge loss plus ℓ_2); the dual adds $0 \le \alpha_i \le C$. A kernel in place of $\mathbf{x}_i^\top \mathbf{x}_j$ gives score $\sum_i \alpha_i y_i k(\mathbf{x}_i, \mathbf{x}) + b$.

6 Selected exercises

Motivation. Two-class softmax, the population target of the logistic loss, and the SVM margin from the dual.

Exercise 4.1 (Two-class softmax is logistic regression)

Let $Q = 2$, scores $s_k = \mathbf{w}_k^\top \mathbf{x}$, labels $y_i = y_{i1} \in \{0,1\}$, $y_{i2} = 1 - y_i$, and $J(\mathbf{w}_1, \mathbf{w}_2)$ the softmax cross-entropy.

(a) Show that $p_1(\mathbf{x}) = \sigma((\mathbf{w}_1 - \mathbf{w}_2)^\top \mathbf{x})$ and $J(\mathbf{w}_1, \mathbf{w}_2) = J_{\log}(\mathbf{w}_1 - \mathbf{w}_2)$, where $J_{\log}$ is the logistic loss of Section 2.
(b) Show that $\nabla_{\mathbf{w}_1} J = -\nabla_{\mathbf{w}_2} J$ and that the Hessian of J in $(\mathbf{w}_1, \mathbf{w}_2)$ is singular.
(c) For $\lambda > 0$ let $(\hat{\mathbf{w}}_1, \hat{\mathbf{w}}_2)$ minimize $J + \frac{\lambda}{2}(\|\mathbf{w}_1\|^2 + \|\mathbf{w}_2\|^2)$. Show that $\hat{\mathbf{w}}_1 = -\hat{\mathbf{w}}_2$ and that $\hat{\mathbf{w}} = \hat{\mathbf{w}}_1 - \hat{\mathbf{w}}_2$ minimizes $J_{\log}(\mathbf{w}) + \frac{\lambda}{4}\|\mathbf{w}\|^2$.

Solution. [Supplement]

(a) Divide by e^{s_1}:

$$p_1 = \frac{e^{s_1}}{e^{s_1} + e^{s_2}} = \frac{1}{1 + e^{-(s_1 - s_2)}} = \sigma((\mathbf{w}_1 - \mathbf{w}_2)^\top \mathbf{x}), \qquad p_2 = 1 - p_1.$$

Hence $J = -\sum_i [y_i \log p_1(\mathbf{x}_i) + (1 - y_i) \log(1 - p_1(\mathbf{x}_i))]$, the logistic loss at $\mathbf{w} = \mathbf{w}_1 - \mathbf{w}_2$.

(b) By (a) and the chain rule, $\nabla_{\mathbf{w}_1} J = \nabla J_{\log}(\mathbf{w}_1 - \mathbf{w}_2)$ and $\nabla_{\mathbf{w}_2} J = -\nabla J_{\log}(\mathbf{w}_1 - \mathbf{w}_2)$. $t \mapsto J(\mathbf{w}_1 + t\mathbf{v}, \mathbf{w}_2 + t\mathbf{v})$ is constant, so $\mathbf{u} = (\mathbf{v}, \mathbf{v}) \neq \mathbf{0}$ has $\mathbf{u}^\top \mathcal{H} \mathbf{u} = 0$; J is convex, $\mathcal{H} \succeq 0$, so $\|\mathcal{H}^{1/2}\mathbf{u}\|^2 = 0$ and $\mathcal{H}\mathbf{u} = \mathbf{0}$.

(c) The penalized objective is strictly convex and coercive; its unique minimizer satisfies

$$\nabla_{\mathbf{w}_1} J + \lambda \hat{\mathbf{w}}_1 = \mathbf{0}, \qquad \nabla_{\mathbf{w}_2} J + \lambda \hat{\mathbf{w}}_2 = \mathbf{0}.$$

Adding and using (b), $\lambda(\hat{\mathbf{w}}_1 + \hat{\mathbf{w}}_2) = \mathbf{0}$, so $\hat{\mathbf{w}}_1 = -\hat{\mathbf{w}}_2 = \hat{\mathbf{w}}/2$. On pairs $(\mathbf{w}/2, -\mathbf{w}/2)$ the objective is $J_{\log}(\mathbf{w}) + \frac{\lambda}{2} \cdot 2\|\mathbf{w}/2\|^2 = J_{\log}(\mathbf{w}) + \frac{\lambda}{4}\|\mathbf{w}\|^2$, and the global minimizer lies there, so $\hat{\mathbf{w}}$ minimizes it. ■

Exercise 4.2 (What logistic regression estimates)

Let $(X, Y) \sim P$ with $Y \in \{\pm 1\}$ and $\eta(\mathbf{x}) = P(Y = +1 \mid X = \mathbf{x}) \in (0, 1)$. For measurable g let $R(g) = \mathbb{E}[\ell(Yg(X))]$ with $\ell(z) = \log(1 + e^{-z})$.
(a) For $\eta \in (0, 1)$ show that $C_\eta(t) = \eta\, \ell(t) + (1 - \eta)\ell(-t)$ is strictly convex with unique minimizer $t^\star = \operatorname{logit}(\eta)$.
(b) Deduce that $g^\star(\mathbf{x}) = \operatorname{logit} \eta(\mathbf{x})$ minimizes R over all measurable g, and that $\operatorname{sign} g^\star$ is the Bayes classifier (predict $+1$ iff $\eta > \frac{1}{2}$).
(c) Show that $C_\eta(t^\star) = -\eta \log \eta - (1 - \eta) \log(1 - \eta)$, so $R(g^\star)$ is the expected binary entropy of $\eta(X)$.

Solution. [Supplement]

(a) $\ell'(z) = \sigma(z) - 1 = -\sigma(-z)$, so $\frac{d}{dt}\ell(-t) = \sigma(t)$ and

$$C_\eta'(t) = -\eta\, \sigma(-t) + (1 - \eta)\sigma(t) = -\eta(1 - \sigma(t)) + (1 - \eta)\sigma(t) = \sigma(t) - \eta.$$

$C_\eta'' = \sigma' > 0$: strictly convex, unique stationary point $\sigma(t) = \eta$, $t = \operatorname{logit}(\eta)$.

(b) $R(g) = \mathbb{E}\, C_{\eta(X)}(g(X)) \geq \mathbb{E}\, C_{\eta(X)}(g^\star(X)) = R(g^\star)$ pointwise by (a); $R(g^\star) \leq \log 2$ is finite and $g^\star$ is measurable. $\operatorname{logit} \eta > 0$ iff $\eta > \frac{1}{2}$.

(c) $\sigma(t^\star) = \eta$ gives $\ell(t^\star) = -\log \eta$, $\ell(-t^\star) = -\log(1 - \eta)$; substitute. The logistic loss estimates the conditional log-odds, so outputs are probabilities. ■

Exercise 4.3 (The margin from the dual variables)

Let the data be linearly separable with both labels present, let $(\mathbf{w}^\star, b^\star)$ solve the hard-margin primal and $\boldsymbol{\alpha}^\star$ the dual, with the KKT conditions of the dual theorem in force.
(a) Show that $\mathbf{w}^\star$ is unique, $\mathbf{w}^\star \neq \mathbf{0}$, and $b^\star$ is uniquely determined.

(b) Show that $\|\mathbf{w}^\star\|^2 = \sum_i \alpha_i^\star$; hence the margin is $\big(\sum_i \alpha_i^\star\big)^{-1/2}$ and the primal and dual values coincide.
(c) Check (b) on the two-point example.

Solution. [Supplement]

(a) *Step 1 (uniqueness of $\mathbf{w}^\star$).* The feasible set is convex (half-spaces). If $(\mathbf{w}_1, b_1), (\mathbf{w}_2, b_2)$ are optimal with $\mathbf{w}_1 \neq \mathbf{w}_2$, the midpoint is feasible and by the parallelogram identity

$$\tfrac{1}{2}\left\|\tfrac{\mathbf{w}_1+\mathbf{w}_2}{2}\right\|^2 = \tfrac{1}{4}\|\mathbf{w}_1\|^2 + \tfrac{1}{4}\|\mathbf{w}_2\|^2 - \tfrac{1}{8}\|\mathbf{w}_1 - \mathbf{w}_2\|^2 < p^\star,$$

a contradiction.

Step 2 ($\mathbf{w}^\star \neq \mathbf{0}$). $\mathbf{w} = \mathbf{0}$ needs $y_i b \geq 1$ for all i, i.e. $b \geq 1$ and $b \leq -1$.

Step 3 ($b^\star$). $\mathbf{w}^\star = \sum_i \alpha_i^\star y_i \mathbf{x}_i \neq \mathbf{0}$, so some $\alpha_i^\star > 0$; complementary slackness and $y_i^2 = 1$ give $b^\star = y_i - \mathbf{w}^{\star\top}\mathbf{x}_i$.

(b) Sum complementary slackness and use $\sum_i \alpha_i^\star y_i = 0$:

$$\sum_i \alpha_i^\star = \sum_i \alpha_i^\star y_i (\mathbf{w}^{\star\top}\mathbf{x}_i + b^\star) = \mathbf{w}^{\star\top}\sum_i \alpha_i^\star y_i \mathbf{x}_i + b^\star \sum_i \alpha_i^\star y_i = \|\mathbf{w}^\star\|^2.$$

Margin $1/\|\mathbf{w}^\star\|$; primal $\tfrac{1}{2}\|\mathbf{w}^\star\|^2$ equals dual $\sum_i \alpha_i^\star - \tfrac{1}{2}\|\mathbf{w}^\star\|^2$.

(c) $\alpha_1 = \alpha_2 = \tfrac{1}{4}$: $\sum_i \alpha_i = \tfrac{1}{2} = \|(1/2, 1/2)\|^2$, margin $\sqrt{2}$. ■